



Indian Institute of Technology
(ISM) Dhanbad

Submitted by
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COURSE

Mining Equipment Reliability, Maintainability & Availability

ASSIGNMENT TITLE

Equipment Selection & Justification:

CAT® 793 Mining Haul Truck

RMA Performance, Failure Analysis & Maintenance Planning

DEPARTMENT

Mining Engineering

SUBMISSION DATE

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SUBMITTED TO

Prof. Ankush Galav



Sources: AEHQ6640-02 (2022) · AEXQ3529-01 (2023) · AEHQ8437-01 (2024) · Cat Performance Handbook Ed.48

CAT®

793

MINING TRUCK

Reliability, Maintainability & Availability (RMA)
Equipment Selection & Justification

229-240 t

Rated Payload

1,976 kW

Gross Power

59.9 km/h

Top Speed



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Introduction & Selection Rationale

The Cat® 793 Mining Truck

The Cat 793 is a 229–240 tonne class mechanical-drive mining truck for demanding open-pit operations. With nearly 5,000 units globally and >21 million C175-16 engine hours, the 793 is one of the most proven haul trucks in the industry (AEXQ3529-01).

Class-Leading Payload

229–240 t NRP — highest for RGMW in class. HP Body saves 3 t tare.

Fuel Efficiency

>10% better than competing trucks; zero fuel burn during retarding.

Autonomous-Ready

500+ autonomous units in service; +30% productivity with Command.



Proven Reliability

>90% mechanical availability; designed for 100,000+ hr service life.

Technology Integration

Standard Product Link Elite (4G/LTE), MineStar™ ready, VIMS.

Total Cat Integration

Engine, transmission, brakes & electronics optimised by one OEM.

Equipment Costs & Economics

Slide 02

~USD \$3.5-5M

Purchase Price
(new unit, ex-works)

USD \$200-350/hr

All-In Operating Cost
(fuel + maint + tyres)

USD \$0.40-0.70/t

Cost per Tonne Hauled
(at optimal utilisation)

>\$1B

Fleet TCO (100-unit fleet
over 20,000 hr life)

Operating Cost Breakdown (per hour, all-in)

Capital & Tyre Cost Item	Estimate
Base Purchase Price (new)	USD \$3.5–5.0M (OEM list, ex-works)
Extended Life Wheel Stations (ELWS)	~USD +\$200–400K option
Body & Hoist Options	~USD +\$80–150K for HP Body upgrade
Autonomous Upgrade (Command-Ready)	Standard; full AHS ~USD +\$300–500K
Tyre Set (6 × 40.00 R57)	~USD \$240,000–360,000 per set
Tyre Life (standard conditions)	~6,000–10,000 hours / set
Tyre Cost per Hour	~USD \$24–60/hr (largest cost item)
Resale / Residual Value (20,000 hr)	~20–30% of purchase price
Cat Reman Rebuild Cost	~30–40% of new unit cost at 20,000 hr

Cost Component	Estimated Cost
Fuel (150–200 L/hr @ USD \$0.60/L)	USD \$90–120/hr
Lubricants & Fluids	USD \$8–15/hr
Tyre Wear & Replacement	USD \$24–60/hr
Planned Maintenance (PM)	USD \$18–35/hr
Major Component Rebuilds	USD \$40–70/hr (amortised)
Operator Labour	USD \$25–50/hr (mine-site rate)
Ground Support (pit crew)	USD \$10–20/hr (allocated)
Overhead & Admin (allocated)	USD \$8–15/hr
TOTAL All-In Operating Cost	USD \$223–385/hr (full TCO)

Key Economic Insight

Tyre cost is the largest variable operating cost (~15–25% of all-in OPEX). AHS Command improves tyre life by +35%, directly reducing cost per tonne. The 793's fuel efficiency advantage (>10% vs. competitors) and near-continuous AHS operation justify the higher capital cost over the fleet life.

Engine Parameter	Specification
Engine Model	Cat® C175-16
Gross Power (SAE J1995)	1,976 kW / 2,650 hp
Net Power (SAE J1349)	1,886 kW / 2,529 hp
Rated Speed	1,750 rpm
Bore × Stroke	175 mm × 220 mm
Displacement	85 L / 5,187 in³
Emissions	Fuel Optimized (Tier 4 Final optional)
Alternator	Brushless, 260 ampere
Starting System	Electric (Air starting optional)
Engine Life (fuel)	1.2M gal (4.5M L) before overhaul

Powertrain Parameter	Specification
Transmission	6-Speed Automatic Powershift (APECS)
Torque Converter	Lock-up (~8 km/h)
Diff. Ratio	1.8:1
Planetary Ratio	16:1
Total Reduction	28.8:1
F1 / F6 (Top Speed)	12.1 km/h / 59.9 km/h
Reverse Speed	11.2 km/h / 7.0 mph
Rear Axle Lubrication	Continuous Lubrication/Filtration
Fleet Hours Logged	>21 million C175-16 hours
Programmable Top Speed	Standard feature

6-Speed Transmission - All Gear Speeds

Gear	km/h	mph
F1	12.1	7.5
F2	16.4	10.2
F3	22.4	13.9

Gear	km/h	mph
F4	30.3	18.8
F5	41.0	25.5
F6	59.9	37.0
Rev	11.2	7.0

Parameter	Value
RGMW – Std. Wheel Stations	386,008 kg / 851,000 lb
RGMW – Ext. Life Wheel Stations	404,000 kg / 890,668 lb
Chassis Weight – Std.	126,410 kg / 278,863 lb
Chassis Weight – ELWS	133,854 kg / 295,098 lb
Body Weight	30,151 kg / 66,471 lb
NRP – Std. Wheel Stations	229 tonnes / 252 tons
NRP – Ext. Life Wheel Stations	240 tonnes / 265 tons
Tyres – Std.	40.00 R57 radial
Tyres – ELWS	46/90 R57 radial
Tyres – Optional	50/80 R57
Steer Angle	36 degrees
Turning Diameter (ISO 7457)	28 m / 93 ft
Braking Standard	ISO 3450:2011
Service Brakes	4-corner, Wet Disc, Oil-Cooled
Parking Brake	4-corner, Multi-disk, Spring Applied
Steering Standard	ISO 5010:2019

Sources: AEHQ6640-02 (2022); AEXQ3529-01 (2023); AEHQ8437-01 (2024)

793 Mining Truck Specifications

Engine

Engine Model	Cat® C175-16	
Gross Power – SAE J1995:2014	1976 kW	2,650 hp
Net Power – SAE J1349:2011	1886 kW	2,529 hp
Rated Speed	1,750 rpm	
Emissions Rating	Fuel Optimized	
Bore	175 mm	6.9 in
Stroke	220 mm	8.7 in
Displacement	85 L	5,187 in³

- Net Power advertised is the power available at the flywheel when the engine is equipped with air intake system, exhaust system, and alternator
- U.S. EPA Tier 4 Final/EU Stage V available for applicable markets

Weights – Approximate

Rated Gross Machine Weight (RGMW)		
Standard Wheel Stations	386 008 kg	851,000 lb
Extended Life Wheel Stations	404 000 kg	890,668 lb
Chassis Weight (CW)		
Standard Wheel Stations	126 410 kg	278,863 lb
Extended Life Wheel Stations	133 854 kg	295,098 lb
Body Weight (BW)	30 151 kg	66,471 lb
Nominal Rated Payload (NRP)		
Standard Wheel Stations	229 tonnes	252 tons
Extended Life Wheel Stations	240 tonnes	265 tons

- Consult your tire manufacturer for maximum tire load
- Chassis weight with full fuel and fluids, standard and mandatory attachments, hoist, body mounting group, rims, and tires

Transmission

Forward 1	12.1 km/h	7.5 mph
Forward 2	16.4 km/h	10.2 mph
Forward 3	22.4 km/h	13.9 mph
Forward 4	30.3 km/h	18.8 mph
Forward 5	41.0 km/h	25.5 mph
Forward 6	59.9 km/h	37 mph
Reverse	11.2 km/h	7 mph
Top Speed – Loaded	59.9 km/h	37 mph

Tires and Rims

40.00 R57
46/90 R57
50/80 R57
737 mm (29 in) Standard rims, 813 mm (32 in) Optional rims
• Quick Change Rims optional
• Extended life wheel stations optional
• Caterpillar recommends the customer evaluate all job conditions and consult tire manufacturer for proper tire selection and TKPH (TMPH) capabilities

Braking System

Service Brakes: Four-Corner, Wet Disc, Oil Cooled, Hydraulically Actuated		
Front Wet Disc Brake Surface Area	89 727 cm²	13,907 in²
Rear Wet Disc Brake Surface Area	134 590 cm²	20,861 in²
Standards (Service and Secondary)	ISO 3450:2011	
Parking Brake: Four-corner, Multi-disk, Spring applied, Hydraulically Released		

Official OEM Technical Specifications Sheet
AEHQ6640-02 (2022) / AEXQ3529-01 (2023)

Brake Disc Surface Areas

- Front Wet Disc: 89,727 cm² / 13,907 in² | Rear: 134,590 cm² / 20,861 in²

Service Braking System

- Type: Four-corner, Wet Disc, Oil-Cooled, Hydraulically Actuated
- Front Wet Disc Area: 89,727 cm² / 13,907 in²
- Rear Wet Disc Area: 134,590 cm² / 20,861 in²
- Standard: ISO 3450:2011
- Parking: 4-corner multi-disk, spring applied, hyd. released

APECS Transmission

- 6-speed Automatic Powershift
- Lock-up torque converter (~8 km/h)
- Body-up Reverse Neutralizer standard
- Programmable top speed
- Controlled Throttle Shifting
- ABS & Dynamic Stability Control (optional)

Steering System

- Steer Angle: 36 degrees
- Turning Diameter: 28 m / 93 ft (ISO 7457:1997)
- Standard: ISO 5010:2019
- Supplemental Steering (automatic) — standard
- Aux. Steering Quick Connect for towing

Final Drives

- Double reduction, planetary, full floating axles
- Differential Ratio: 1.8:1
- Planetary Ratio: 16:1
- Total Reduction: 28.8:1
- Rear Axle Continuous Lubrication/Filtration — std.

Retarding System

- 4-corner mechanical resistive retarding
- Zero fuel burn during retarding
- ARC (Automatic Retarder Control) — adjustable
- ARC allows 3–6% higher descent speeds
- Additional Retarding option for steep hauls
- Engine Overspeed Protection standard

Tyres & Rims

- Std: 40.00 R57 (Standard Wheel Stations)
- ELWS: 46/90 R57 (Extended Life WS)
- Optional: 50/80 R57
- Std Rims: 737 mm (29 in); Opt: 813 mm (32 in)
- Quick Change Rims optional
- TPMS2 Payload Monitoring — standard

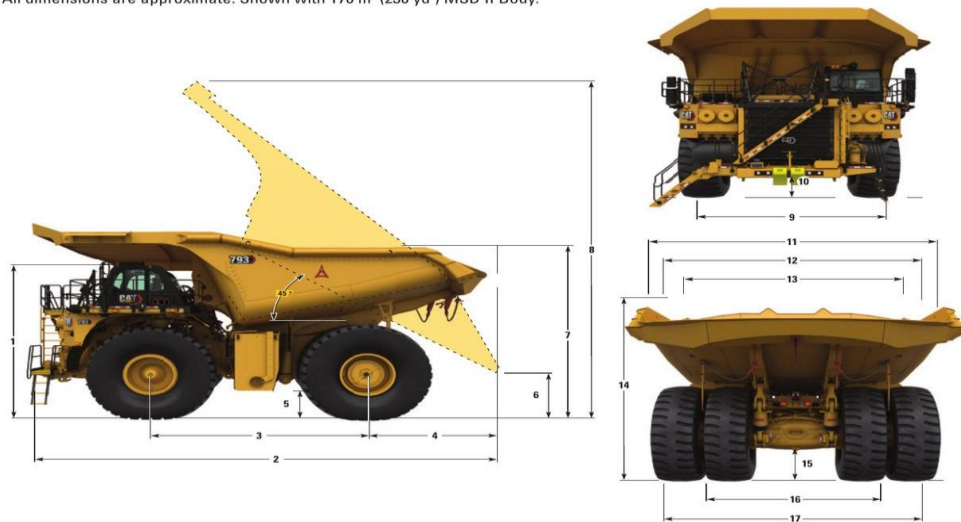
Dimensions

Slide 06

Shown with 176 m³ (230 yd³) MSD II Body | All dimensions approximate

Dimensions

All dimensions are approximate. Shown with 176 m³ (230 yd³) MSD II Body.



OEM Dimensions Diagram — AEHQ6640-02 (2022)

Dimension Reference	Measurement
① Height to ROPS	5,712 mm (18 ft 9 in)
② Overall Length	13,360 mm (44 ft 10 in)
③ Wheelbase	5,905 mm (19 ft 5 in)
④ Rear Axle to Tail	3,997 mm (13 ft 2 in)
⑤ Ground Clearance	1,041 mm (3 ft 5 in)
⑥ Dump Clearance	1,301 mm (5 ft 4 in)
⑦ Loading Ht (Empty)	6,181 mm (20 ft 4 in)
⑧ Ht – Body Raised	13,710 mm (45 ft 0 in)
⑨ CL Front Tire Width	5,609 mm (18 ft 5 in)
⑩ Overall Canopy Width	8,563 mm (28 ft 1 in)
⑪ Outside Body Width	7,942 mm (26 ft 1 in)
⑫ Inside Body Width	7,334 mm (24 ft 1 in)
⑬ Front Canopy Height	7,074 mm (23 ft 3 in)
⑭ CL Rear Dual Tire	4,963 mm (16 ft 3 in)
⑮ Overall Tire Width	7,605 mm (24 ft 11 in)

Performance: Gradeability & Retarding

Slide 07

gear, then down to maximum speed. Usable rimpull will depend upon traction available and weight on drive wheels. Chart utilizes extended life wheels, stations and 50/80R57 tires.

Typical Field Empty Weight

Gross Machine Operating Weight
404 000 kg (890,669 lb)

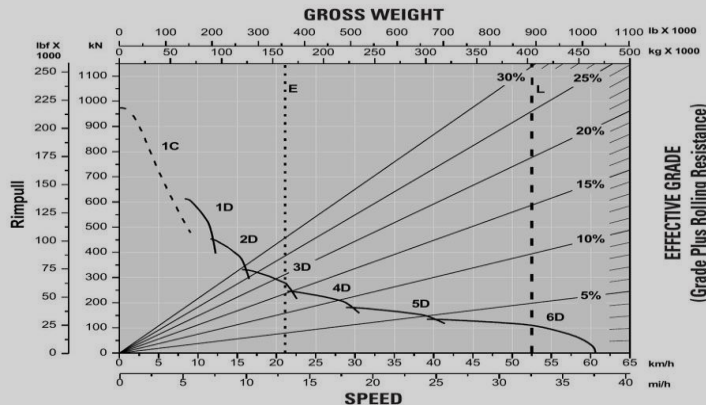
- 1 – 1st Gear
- 2 – 2nd Gear
- 3 – 3rd Gear
- 4 – 4th Gear
- 5 – 5th Gear
- 6 – 6th Gear

E – Empty
L – Loaded

* at sea level

Torque Converter Drive

Direct Drive



Gradeability / Speed / Rimpull Chart — AEHQ6640-02
(ELWS with 50/80R57 tyres, sea level)

Speed Summary

- Top Speed (F6, Loaded): 59.9 km/h / 37.0 mph
- Advertised Top Speed: 60 kph / 37 mph
- Reverse Speed: 11.2 km/h / 7.0 mph

Gradeability

- Max grade capability: up to 28% loaded
- Class-leading speed on 10% grade
- 6-gear APECS maximises torque on grade

Retarding Performance

- 4-corner wet disc retarding (oil-cooled)
- ARC allows 3–6% higher descent speed
- Zero fuel burn during mechanical retarding
- Additional Retarding option for steep hauls

Body, Hoist & Suspension

Slide 08

Body & Hoist	Value
Body Type	HP Body (standard); MSD II optional
Struck Capacity	112.6–151 m³ / 147–197.5 yd³
Heaped (SAE 2:1)	162–190 m³ / 212–249 yd³
Hoist Type	Twin 2-stage hydraulic cylinders
Pump Flow (High Idle)	819 L/min / 216 gal/min
Relief Valve Setting	20,370 kPa / 2,955 psi
Body Raise Time	22 seconds (high idle)
Body Lower Time	23 seconds (float)
Body Power Down	20.7 seconds (high idle)

Suspension & Hydraulics	Value
Suspension Type	Self-contained N ₂ /oil cylinders
Mounting	Pin-to-pin, double shear clevis
Front Stroke	317 mm / 12.48 in
Rear Stroke	165 mm / 6.50 in
Rear Axle Oscillation	± 4.9 degrees
Brake/Hoist Tank	1,315 L / 347 gal
Hydraulic Filters	1,000 hr standard
Snubbing Valve	Standard on hoist cylinders
AutoHoist	Optional — auto raise & engine speed

Body & Hoist Key Features

Low-Profile Design

Reduces centre of gravity; improves stability on grade and corners.

High-Strength Steel

AR steel liners in floor/sides; wear strips standard fitment.

Canopy Protection

Provides rock protection for cab and ROPS structure.

Twin Hoist Cylinders

Snubbing valve prevents body slamming; auto lower in float.

Body Heat Option

Prevents wet material freeze-adhesion in cold climates.

Fluid System	Capacity
Fuel Tank (Standard)	2,839 L / 750 gal
Fuel Tank (Optional)	4,922 L / 1,300 gal
Diesel Exhaust Fluid (DEF)	235 L / 62 gal
Cooling System	1,074 L / 284 gal
Crankcase	312 L / 82 gal
Front Wheels, Each	30 L / 8 gal
Differential & Final Drives	984 L / 260 gal
Steering System (incl. tank)	290 L / 77 gal
Brake/Hoist System (incl. tank)	1,315 L / 347 gal
Torque Converter Sump	102 L / 27 gal
TC/Transmission System	209 L / 55 gal

Weight Distributions - Approximate

Condition	Front Axle	Rear Axle
Empty	51%	49%
Loaded	33%	67%

Weight distribution notes:

- Optimised for Cat body — distributions verified with Cat HP Body.
- Rear-biased load (67%) ensures optimal traction at rear drive wheels.
- Front-heavy empty (51%) improves stability and steering response.
- Consult tyre manufacturer for maximum per-axle tyre load limits.
- ELWS allows 240 t payload vs. 229 t Std., with matching RGMW.
- Weight distributions are approximate and vary with body options.

Cab Climate Ratings

- AC: 9.3 kW / 31,733 BTU/hr (to 50°C / 122°F) | Heater: 10.7 kW / 36,510 BTU/hr (to -30°C / -22°F)

Key Features & Benefits

Slide 10

Fuel Efficiency

>10% more fuel-efficient vs. competitive trucks. Zero fuel burn during retarding. APECS enables higher gear on grade.

Cycle Time

Up to 10% faster cycle times. Class-leading speed on 10% grade. Lock-up torque converter improves acceleration.

Payload Advantage

Up to 6% greater payload vs. competitors at rated GMW. HP Body saves additional 3 tonnes tare weight.

APECS Powertrain

More continuous torque/rimpull to ground. ARC allows 3–6% higher retarding speeds reducing cycle time & tyre wear.

Cab Environment

74 dB(A) quiet cab. Automated temperature control. HEPA filtration (96% dust reduction). 360° surround view optional.

Connectivity

Fast Ethernet (250 channels, 10 Mbps). Product Link Elite (4G/LTE). Real-time decision making. Remote Troubleshoot.

Safety Systems

ABS + DSC optional. Enhanced Traction Control. Machine Speed Limiting. Driver Safety System fatigue alerts standard.

Serviceability

Modular HVAC, modular radiator, AutoLube std. 1,000-hr hydraulic filters. Electronic O&M Manual standard.

Autonomy Ready

Command-Ready Option standard. 500+ autonomous 793s in service. +30% productivity, +11% fuel efficiency with AHS.

ROPS & FOPS Certification

ISO 3471:2008 (Operator); ISO 13459:2012 (Trainer) for ROPS. FOPS: ISO 3449:2005 Level II for Operator; ISO 13459:2012 Level II for Trainer.

Anti-lock Braking (ABS)

Optional ABS prevents wheel lockup, maintains steerability during emergency braking on slippery or uneven surfaces.

Dynamic Stability Control

DSC (optional) reduces rollover and loss-of-control risk. Works with ARC and engine management systems.

Traction Control System

Prevents excessive wheel spin on soft or wet surfaces. Integrates with APECS to maximise traction.

Automatic Retarder Control

ARC senses grade and payload; sets retarding speed automatically. Prevents brake overheating on long descents.

Driver Safety System

Optional DSS provides real-time fatigue and distraction alerts. Reduces operator-error incidents on long shifts.

360° Surround View Camera

Optional 360° camera system gives operator full visibility around the truck. Detects personnel and obstacles.

Object Detection (Detect)

Standard Integrated Object Detection using radar + camera warns of hazards in the immediate vicinity.

Emergency Shutdown System

Standard at front bumper and in-cab. Ground Level Engine Shutdown and Start Lockout both standard fitment.

Connectivity (Standard)

- Product Link Elite – 4G/LTE cellular standard
- Dual-mode 4G/LTE + satellite optional
- Fast Ethernet: 250 channels, 10 Mbps @ 1 hz
- Remote Flash — OTA software updates standard
- Remote Troubleshoot: –50% software downtime
- Fault repair delay decreased by 10%

VIMS & Diagnostics (Standard)

- Vital Information Management System (VIMS)
- VIMS Message Center with Advisor
- Interactive touchscreen vehicle info display
- Road Analysis Control (RAC) — standard
- Cyber Security — standard
- Electronic O&M Manual — standard
- Ground Level VIMS Data Port — standard

MineStar™ Solutions

- Fleet — fleet management & dispatch
- Terrain — grade control & reporting
- Detect — proximity & object detection
- Health — predictive maintenance & alerts
- Command — full autonomous haulage
- Command-Ready Option: standard fitment
- Equipment Care Advisor (ECA) optional

Technology Feature	Description / Benefit
TPMS2 Payload Monitoring	Double accuracy vs. previous — standard fitment. Improves loading accuracy and tyre management.
APECS (Advanced Power Electronic Control Strategy)	Optimises gear selection, TC lock-up, and retarding for fuel efficiency and productivity.
ARC (Automatic Retarder Control)	Senses grade and payload; sets retarding speed automatically. Reduces operator workload, prevents overheating.
Operator / Speed Coaching	Real-time operator feedback on speed, gear, and braking to maximise productivity and fuel economy.
Road Analysis Control (RAC)	Monitors and reports haul road conditions for maintenance prioritisation and tyre/suspension protection.

Autonomous Haulage (MineStar™ Command)

Slide 13

500+

Autonomous 793s in service globally (AEXQ3529-01)

+30%

Productivity boost vs. staffed operation

+11%

Fuel efficiency vs. staffed trucks

+35%

Increase in useful tyre life

~5B t

Tonnes hauled by Command-equipped trucks

-50%

Software downtime via Remote Troubleshoot

Command for Hauling Features

- Fully integrated factory offering — not an aftermarket add-on
- Near-continuous utilisation (reduced shift-change downtime)
- Consistent speed & gear selection eliminates operator variability
- Integrated with MineStar Fleet for dispatch & route optimisation
- Command-Ready Option is standard on all new 793 trucks

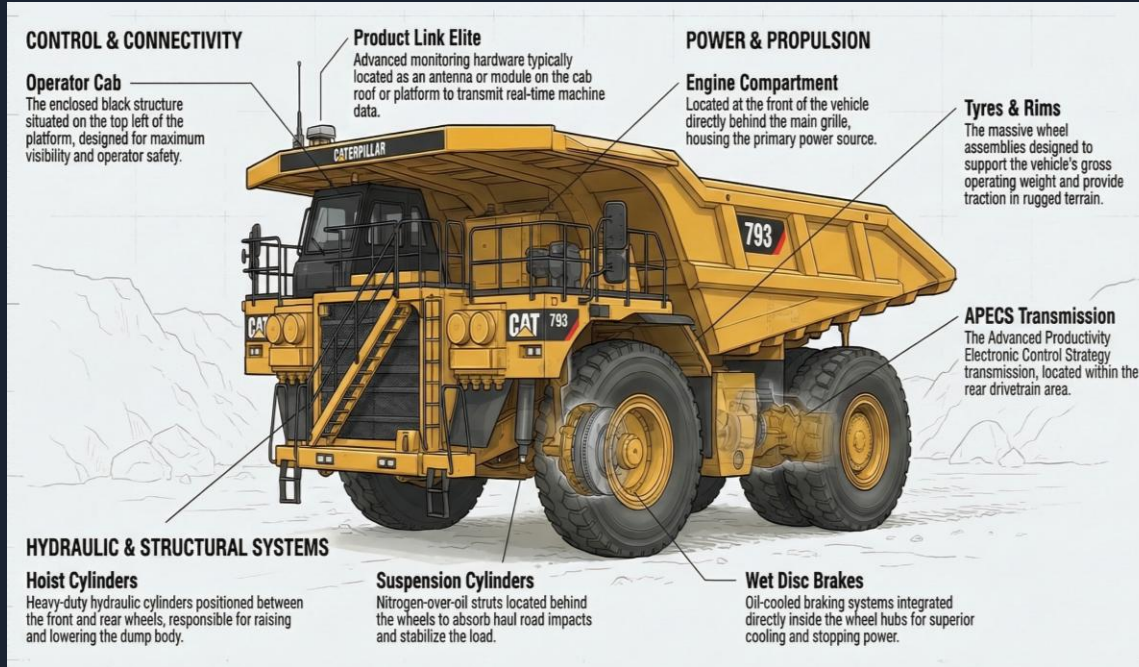
Key Benefits of AHS

- Safety: removes operator from high-risk haul-road environment
- Productivity: +30% over staffed operation (OEM benchmark)
- Fuel: +11% efficiency (consistent throttle/braking vs. human)
- Tyres: +35% useful tyre life (smooth acceleration/decel profile)
- Availability: near-continuous operation; no fatigue breaks

Component Schematic - CAT 793

Slide 14

Key sub-systems and their locations on the truck



- 1 C175-16 Engine**
1,976 kW, 16-cyl, 85 L
- 2 APECS Transmission**
6-speed auto powershift
- 3 Body Hoist Cylinders**
Twin 2-stage hydraulic
- 4 Wet Disc Brakes**
4-corner oil-cooled
- 5 Suspension Cylinders**
N₂/oil, F:317mm, R:165mm
- 6 ROPS Cab**
ISO 3471/3449 certified
- 7 Tyres & Rims**
40.00 R57 or 46/90 R57
- 8 Product Link Elite**
4G/LTE, MineStar ready

Grade & Haul Road Requirements

- Max grade: up to 28% loaded (performance charts)
- Class-leading speed on 10% grade (APECS F1–F2)
- ARC allows 3–6% faster descent vs. competitors
- Turning diameter 28 m (ISO 7457); steer angle 36°
- Min. haul road width: $\sim 3 \times$ overall tyre width ≈ 22.8 m
- Traction Control prevents wheel spin on wet/soft surfaces

Shovel / Loader Match

- Optimal 3–4 pass: Cat 6040/6060 HMS (hydraulic mining shovel)
- 3–4 pass match: Cat 7295/7395 ERS (electric rope shovel)
- 5-pass match: Cat 995 High-lift wheel loader
- Target 3–5 pass loading for optimal cycle time and tyre wear
- TPMS2 payload monitoring ensures accurate loading targets
- HP Body: 162–190 m³ heaped for wide loading flexibility

Climate & Environment

- Cab AC rated to +50°C (122°F) ambient
- Heater to –30°C (–22°F); optional antifreeze to –50°C
- Optional coolant/oil heater for cold-weather starts
- Optional HEPA filtration: 96% respirable dust reduction
- Extended life coolant to –35°C standard
- High-altitude torque converter optional

Operational Suitability

- Mechanical availability consistently >90% (OEM field data)
- Suitable for 24/7 continuous operation; 100,000+ hr design life
- Autonomous-ready (Command-Ready Option standard fitment)
- ~5,000 units in global operation — proven OEM support network
- Remote Troubleshoot reduces software downtime by 50%
- Cat Reman programme supports mid-life rebuild at ~20,000 hrs

Common Failure Modes & Reliability Analysis

Slide 16

Sub-system	Failure Mode	Probability	Root Cause	Impact	Mitigation / Monitoring
Tyre & Rim System	Tyre overloading / overheating (TKPH exceedance)	High	TKPH non-compliance, poor roads	Blowout risk; unplanned downtime	TPMS2; TKPH compliance; haul road PM
Wet-Disc Braking	Brake oil overheating / oil film breakdown	Med-High	Prolonged retarding on steep grades	Loss of braking effectiveness	ARC control; brake oil temp via VIMS
Engine & Fuel System	Injector fouling / abnormal wear	Medium	Fuel contamination; extended intervals	Power loss; high fuel consumption	S-O-S (500 hr); water-sep. fuel filter
Suspension Cylinders	N ₂ charge loss / seal failure	Medium	Overloading; rough roads; wear cycles	Rough ride; tyre & frame overloading	Periodic recharge/rebuild; weight checks
APECS Transmission	Clutch pack wear / oil pump degradation	Low-Med	High cycles; contaminated fluid	Rough shifting; torque loss	1,000 hr filter; S-O-S fluid analysis
Hydraulic Hoist	Hoist cylinder seal failure / relief valve wear	Low	Contamination; high cycle count	Slow body raise; body drift	1,000 hr hydraulic filter; valve check

RMA Benchmark Data (OEM Field Reports)

Interval	Activity
Every 10 hrs	Walk-around inspection, tyre pressure, fluid levels, lights, alarms & brakes
Every 250 hrs	Engine oil & filter, hydraulic breathers, battery check
Every 500 hrs	Fuel filters, differential/final drive oil sample (Cat S-O-S analysis)
Every 1,000 hrs	Transmission & converter filter, steering filter, brake wear indicator check
Every 1,000 hrs	Standard hydraulic filters (optional 2,000-hr filters available)
Every 2,000 hrs	APECS & ARC calibration, VIMS check, suspension cylinder inspection
Every 6,000 hrs	Engine top-end inspection, suspension cylinder rebuild, alternator service
At ~20,000 hrs	Mid-life rebuild: engine, transmission, final drives rebuilt to like-new spec
100,000+ hrs	Design life — built to be rebuilt multiple times (Cat Reman programme)

Metric	Value	Source
Mechanical Availability	>90% (consistent)	AEHQ8437-01
MTBF (comparative)	Industry-leading	Gustafson 2013
MTTR	Reduced (modular)	AEHQ8437-01
AHS Productivity	+30% vs. staffed	AEHQ8437-01
AHS Fuel Efficiency	+11% vs. staffed	AEXQ3529-01
AHS Tyre Life	+35% useful life	AEXQ3529-01
Remote Diag. Downtime	–50% software	AEHQ8437-01
Fault Repair Delay	–10%	AEHQ8437-01
Design Service Life	100,000+ hours	AEHQ8437-01
Engine Overhaul Life	1.2M gal (4.5M L)	AEXQ3529-01
C175-16 Fleet Hours	>21 million hrs	AEXQ3529-01
Units in Operation	~5,000 globally	AEHQ8437-01

[1] Caterpillar Inc. (2022). 793 Mining Truck – Technical Specifications (AEHQ6640-02, Build 08A). © 2022 Caterpillar.

[2] Caterpillar Inc. (2023). Cat® 793 Mining Truck – Key Features & Technical Specifications (AEXQ3529-01). © 2023 Caterpillar.

[3] Caterpillar Inc. (2024). 793 Mining Truck Brochure (AEHQ8437-01). © 2024 Caterpillar Inc., Peoria, IL.

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